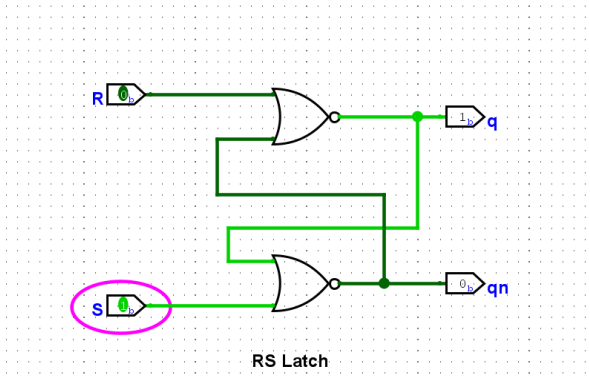
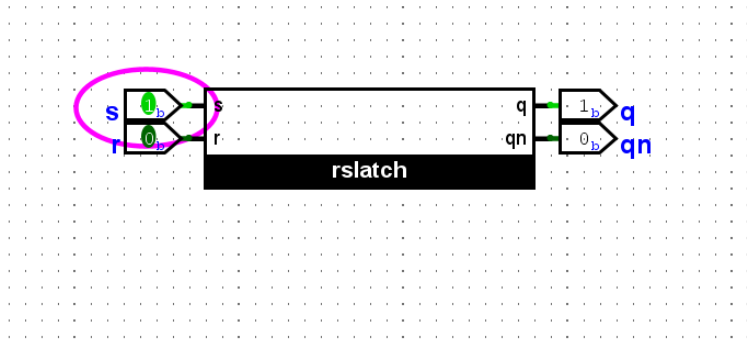
VHDL Assignment 3

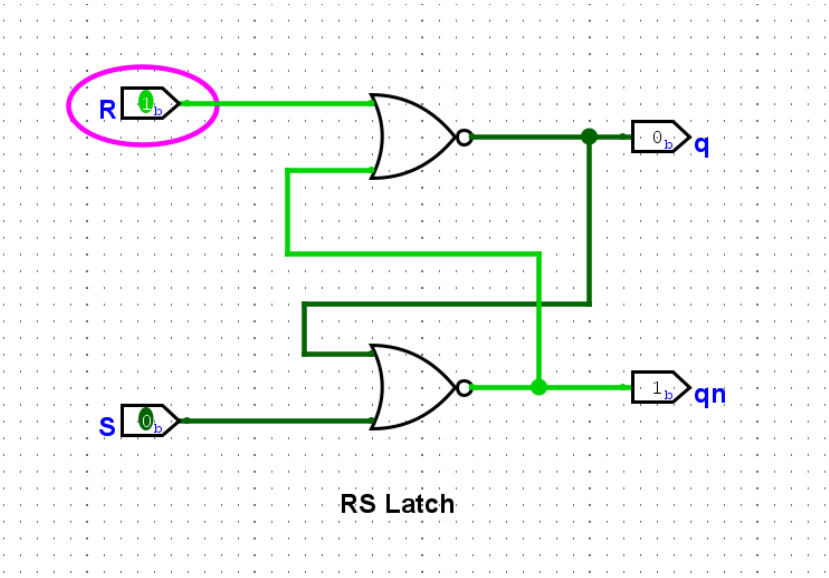
2023008568 이재원 제출일: 2026.11.13

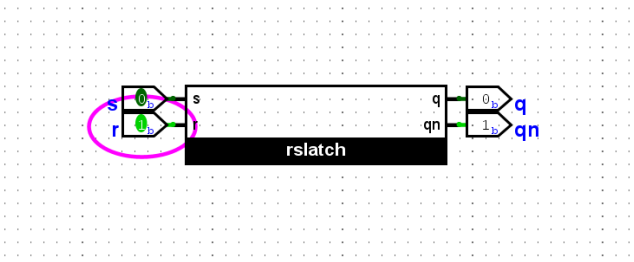
사용 도구: Logisim-evolution 5.0.0, Questa Altera Starter FPGA Edition 2025.2

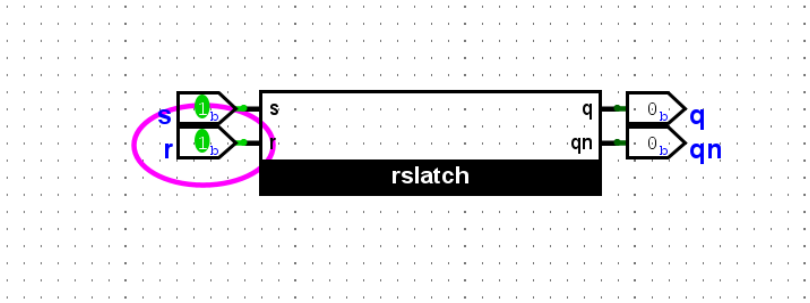


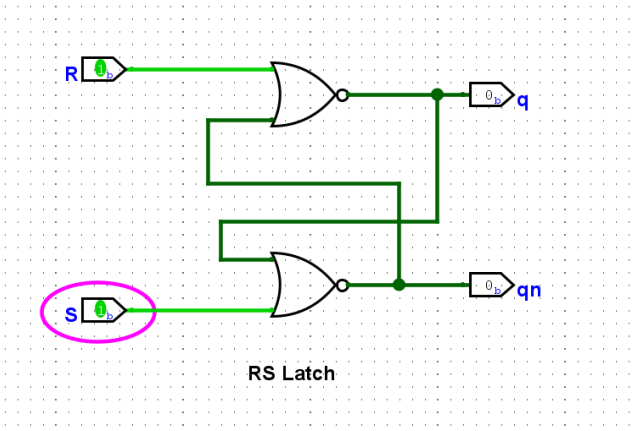
1. RS LATCH






s
r
0b
0b
s
r
rslatch
q
qn
0b
1b
q
qn


s
r
1b
1b
s
r
rslatch
q
qn
0b
0b
q
qn



RS Latch

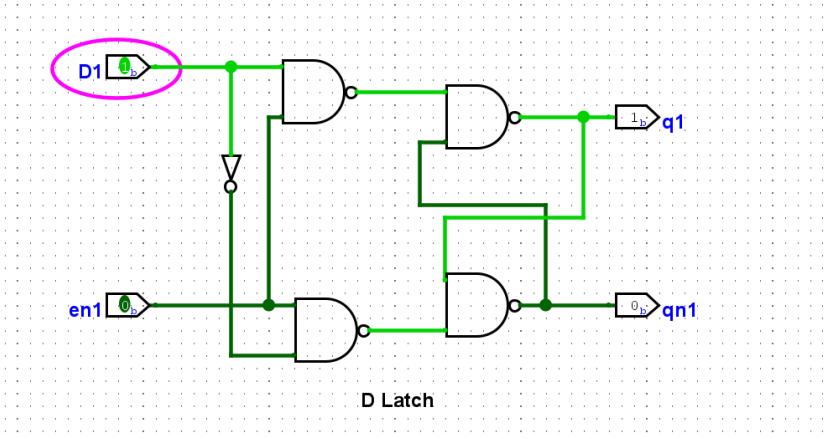
```vhdl
library ieee;
use ieee.std_logic_1164.all;

entity rslatch is
    port (
        s  : in  std_logic;
        r  : in  std_logic;
        q  : out std_logic;
        qn : out std_logic
    );
end entity rslatch;

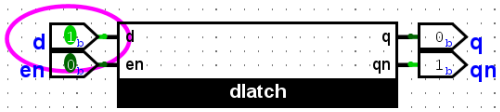
architecture behavioral of rslatch is
    signal q_i  : std_logic := '0';
    signal qn_i : std_logic := '1';
begin
    process (s, r)
    begin
        if s = '1' and r = '0' then
            q_i  <= '1';
            qn_i <= '0';
        elsif s = '0' and r = '1' then
            q_i  <= '0';
            qn_i <= '1';
        elsif s = '1' and r = '1' then
            q_i  <= '0';    -- forbidden: NOR latch gives 0, 0
            qn_i <= '0';
        end if;
        -- s = '0' and r = '0': hold (no assignment)
    end process;

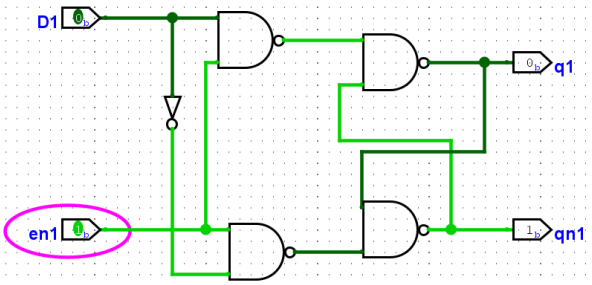
    q  <= q_i;
    qn <= qn_i;
end architecture behavioral;
```

2. D LATCH



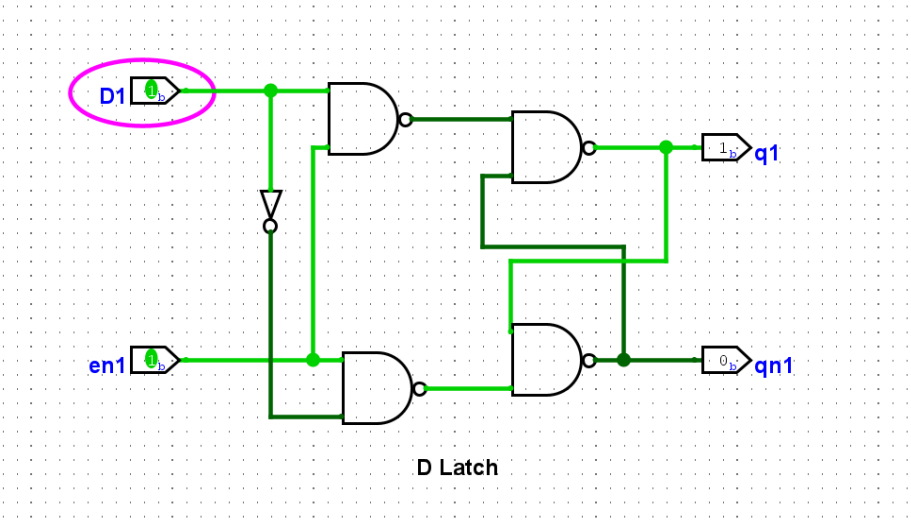
D Latch





D Latch





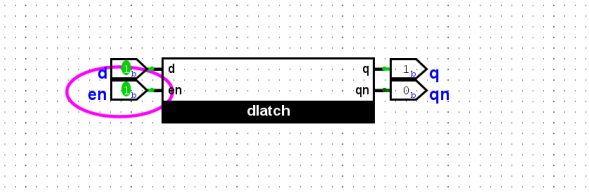
D Latch

```vhdl
library ieee;
use ieee.std_logic_1164.all;

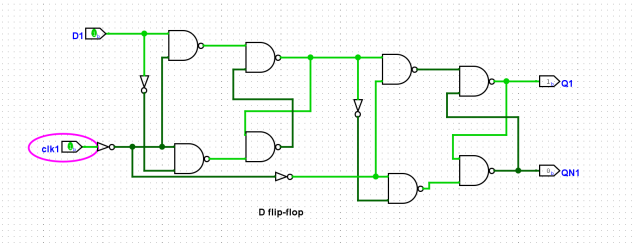
entity dlatch is
    port (
        d  : in  std_logic;
        en : in  std_logic;
        q  : out std_logic;
        qn : out std_logic
    );
end entity dlatch;

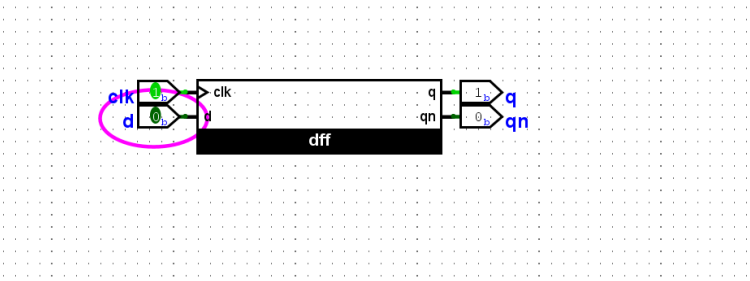
architecture behavioral of dlatch is
    signal q_i : std_logic := '0';
begin
    process (d, en)
    begin
        if en = '1' then
            q_i <= d;
        end if;
        -- en = '0': hold (no assignment)
    end process;

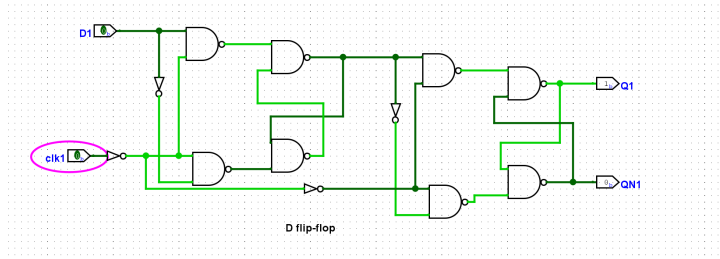
    q  <= q_i;
    qn <= not q_i;
end architecture behavioral;
```

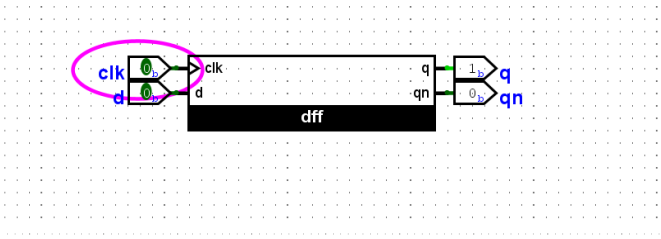


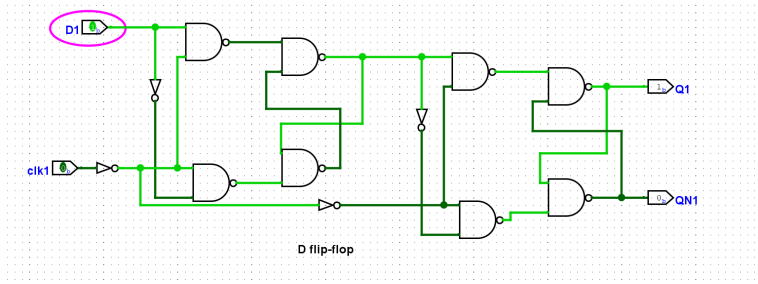
3. D Flip-Flop

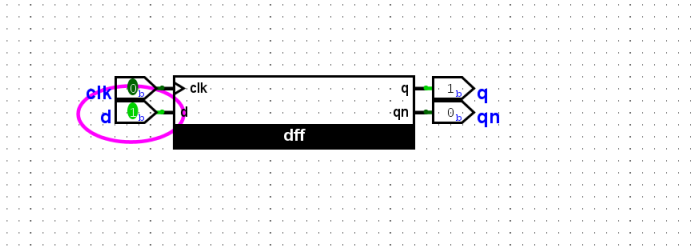

D1
clk1
Q1
QN1
D flip-flop


clk
d
clk
d
q
qn
dff
q
qn


D1
clk1
Q1
QN1
D flip-flop


clk
d
clk
d
q
qn
dff
q
qn


D1
clk1
Q1
QN1
D flip-flop



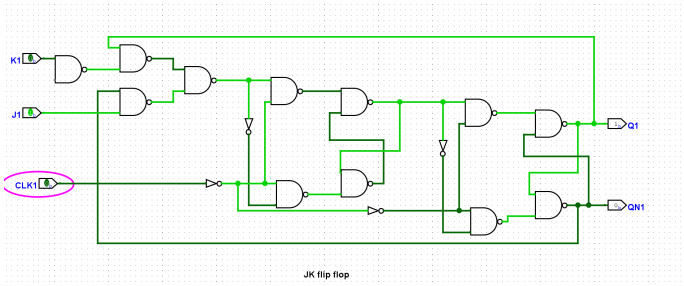

```vhdl
library ieee;
use ieee.std_logic_1164.all;

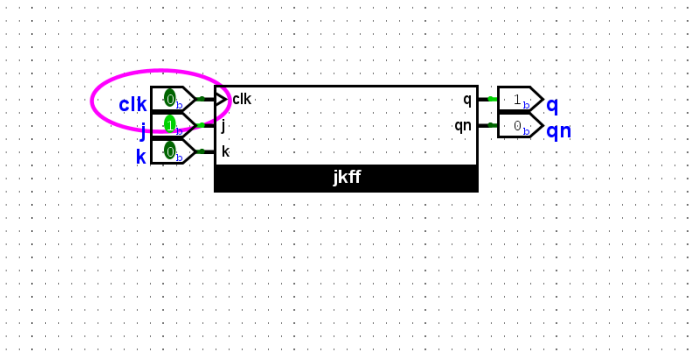
entity dff is
    port (
        clk : in  std_logic;
        d   : in  std_logic;
        q   : out std_logic;
        qn  : out std_logic
    );
end entity dff;

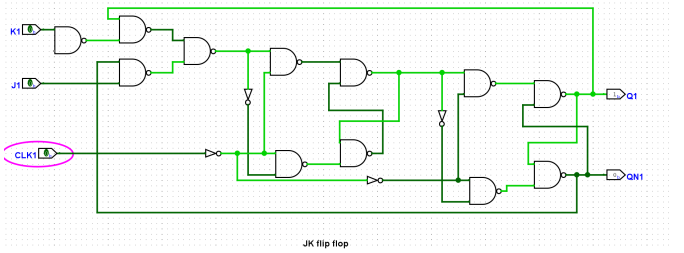
architecture behavioral of dff is
    signal q_i : std_logic := '0';
begin
    process (clk)
    begin
        if rising_edge(clk) then
            q_i <= d;
        end if;
    end process;

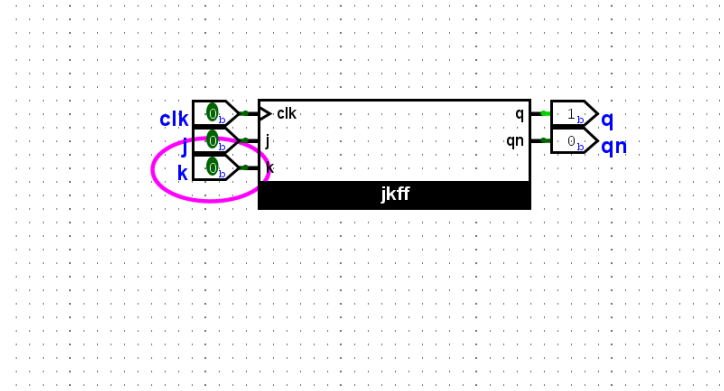
    q  <= q_i;
    qn <= not q_i;
end architecture behavioral;
```

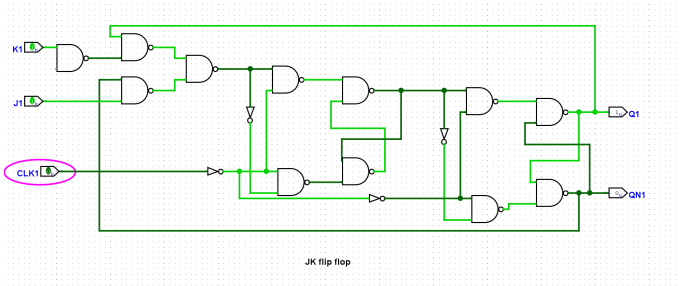
4. JK Flip-Flop

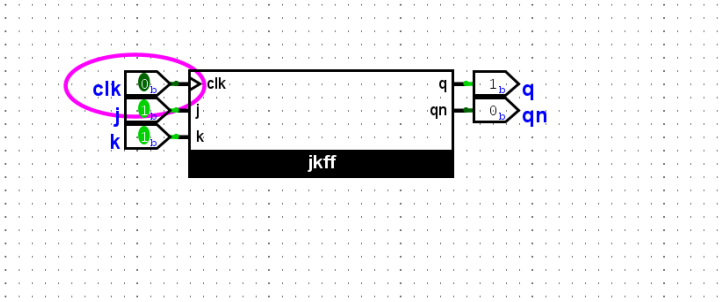


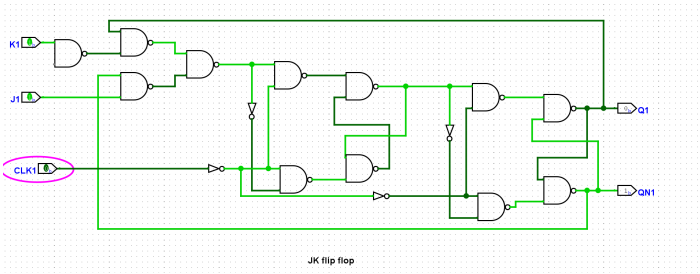




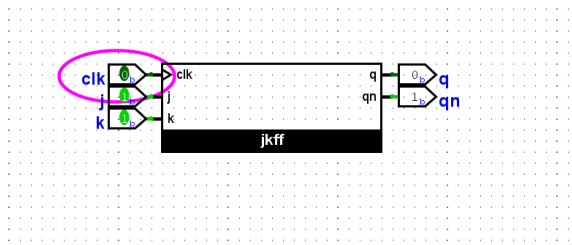








JK flip flop



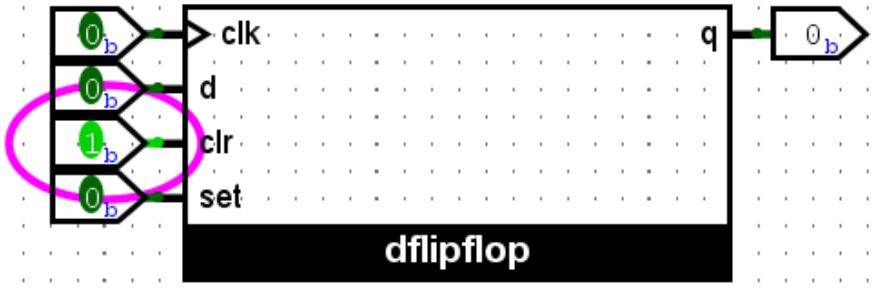

```vhdl
library ieee;
use ieee.std_logic_1164.all;

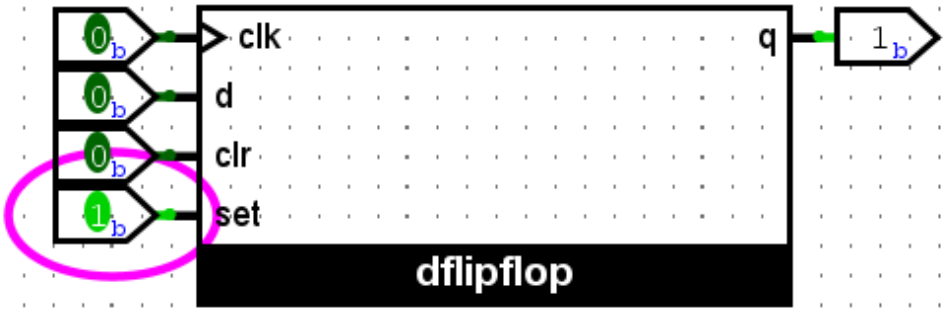
entity jkff is
    port (
        clk : in  std_logic;
        j   : in  std_logic;
        k   : in  std_logic;
        q   : out std_logic;
        qn  : out std_logic
    );
end entity jkff;

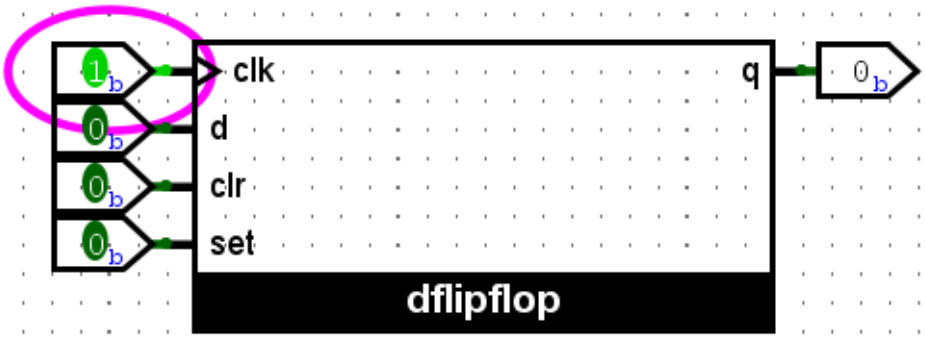
architecture behavioral of jkff is
    signal q_i : std_logic := '0';
begin
    process (clk)
    begin
        if rising_edge(clk) then
            if j = '0' and k = '0' then
                q_i <= q_i;         -- Hold
            elsif j = '0' and k = '1' then
                q_i <= '0';         -- Reset
            elsif j = '1' and k = '0' then
                q_i <= '1';         -- Set
            else
                q_i <= not q_i;     -- Toggle
            end if;
        end if;
    end process;

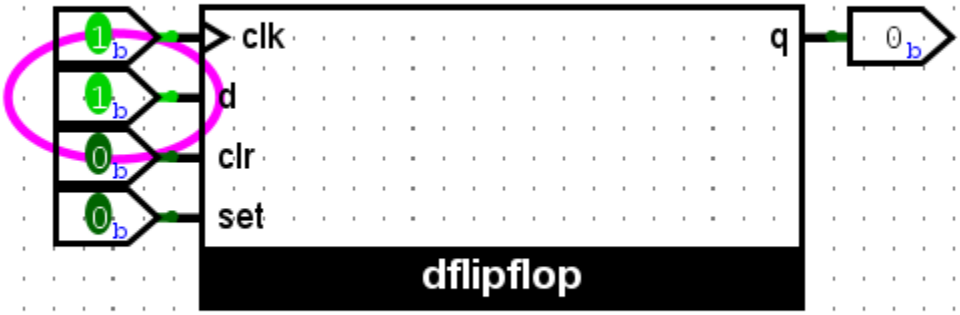
    q  <= q_i;
    qn <= not q_i;
end architecture behavioral;
```

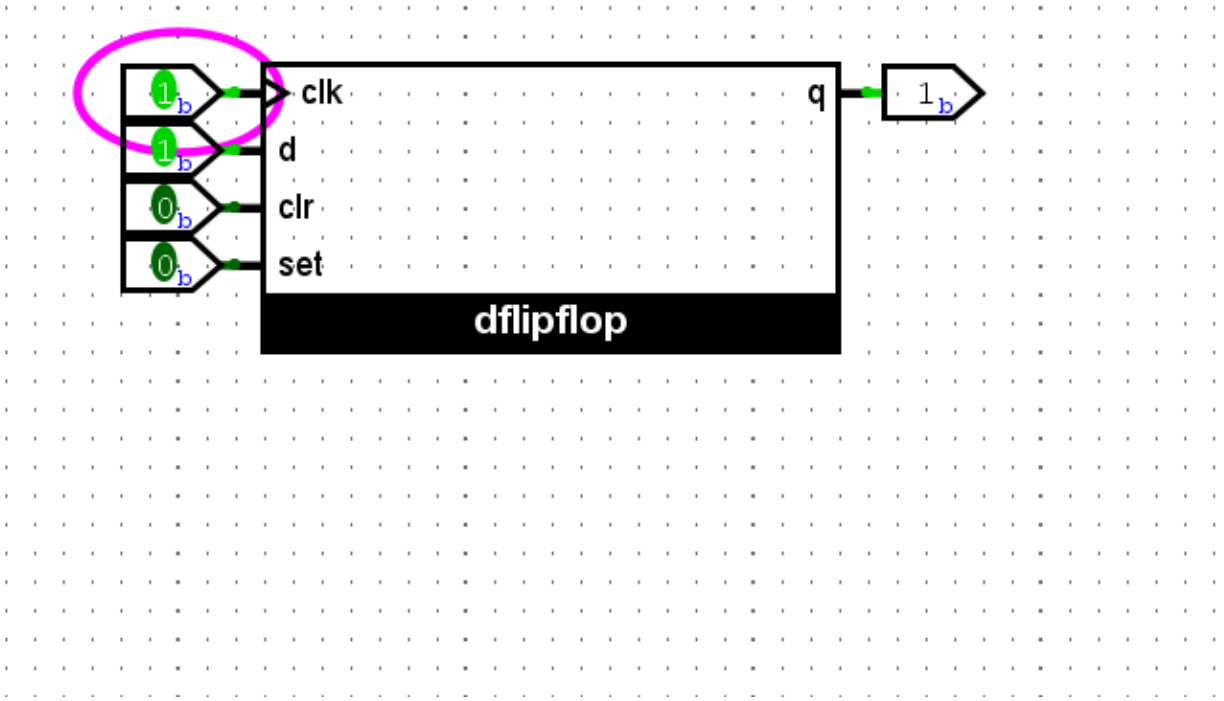
5. D Flip-Flop with Asynchronous Set and Clear

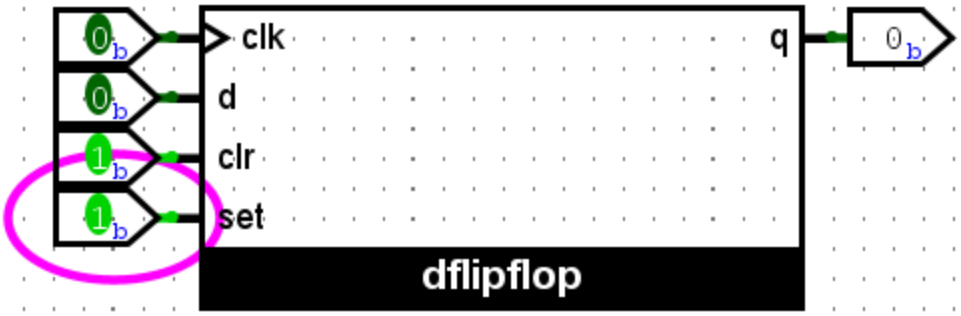





1b
0b
0b
0b
clk
d
clr
set
dflipflop
q
0b


1b
1b
0b
0b
clk
d
clr
set
dflipflop
q
0b


1b
1b
0b
0b
clk
d
clr
set
q
1b
dflipflop


0b
0b
1b
1b
clk
d
clr
set
dflipflop
q
0b

```vhdl
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity dflipflop is
    port (
        clk : in  std_logic;  -- Clock signal
        d   : in  std_logic;  -- Data input
        clr : in  std_logic;  -- Asynchronous clear signal
        set : in  std_logic;  -- Asynchronous set signal
        q   : out std_logic   -- Data output
    );
end dflipflop;

architecture bhv of dflipflop is
begin
    process (clk, clr, set)
    begin
        -- Asynchronous clear has the highest priority
        if clr = '1' then
            q <= '0';
        -- Asynchronous set has the second-highest priority
        elsif set = '1' then
            q <= '1';
        -- Synchronous clock-based behavior
        elsif (clk'event and clk = '1') then
            q <= d;
        end if;
    end process;
end bhv;
```